Tutorial 4 - Digital Systems

1. Consider a logic circuit with three inputs, namely x, y, and z, and three outputs, namely A, B, and C. When the binary input (i.e., xyz) is 0, 1, 2, or 3, the binary output (ABC) should be one greater than the input. For example, if $xyz = 000$, then $ABC = 001$. When the binary input is 4,5,6, or 7, the binary output should be one less than the input. Design a circuit to realize this using only NAND gates.

2. A half-adder takes two inputs (each of which is a single bit number) and produces the sum and carry of addition of the two input bits. In this question, we explore the possibility of using a half-adder as a primitive. Design a 2-input OR gate using minimum number of half adders.

3. For a 4-bit ripple carry adder (as shown in Figure 1), compute the sum and carry at each stage for the binary inputs given by:

$$a_3a_2a_1a_0 = 1011_2, \quad b_3b_2b_1b_0 = 1101_2$$

Q3 solved here itself

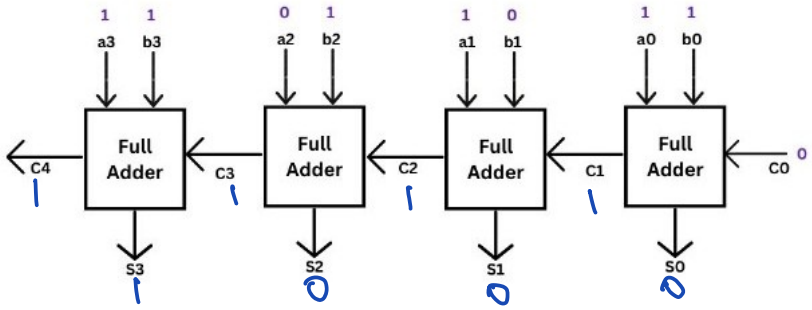


Figure 1: 4-bit Ripple Carry Adder

4. Consider the same binary inputs as in question 3. Assume that we have a 4-bit carry lookahead adder (instead of a ripple carry adder). Perform the addition showing the numerical values of the generate and propagate signals as well as the carry calculations using the generate and propagate equations.

5. Design a subtractor unit using a 4-bit comparator, a 4-bit adder and some additional gates to perform $A - B$ if $A > B$ and $B - A$ otherwise. Assume A and B are two 4-bit numbers.

6. Design an adder/subtractor unit using a 4-bit binary adder and some external gates. The adder/subtractor should give out $A + B$ if $M = 0$ and $A - B$ if $M = 1$ where M is a mode bit. Also, provide an indicator for checking the overflow.

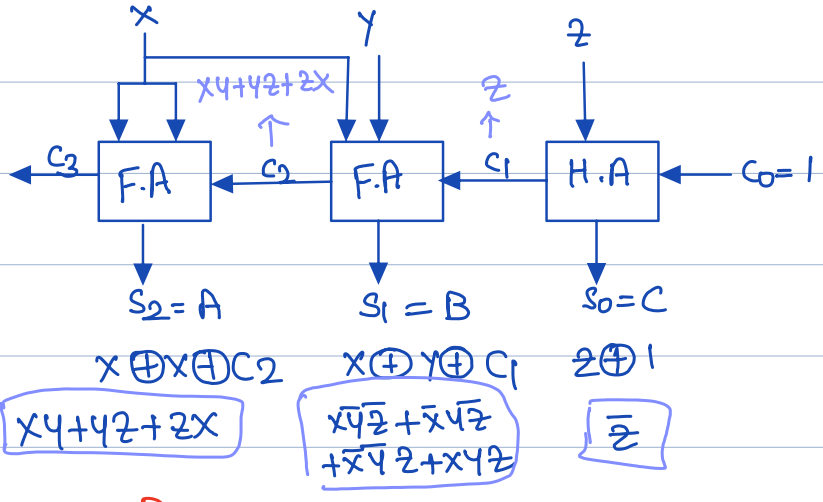
7. You are given a 2:4 decoder, a 2-input OR gate, and a 3-input OR gate. Using these components, design a circuit that takes A and B as inputs and generates the following four outputs:

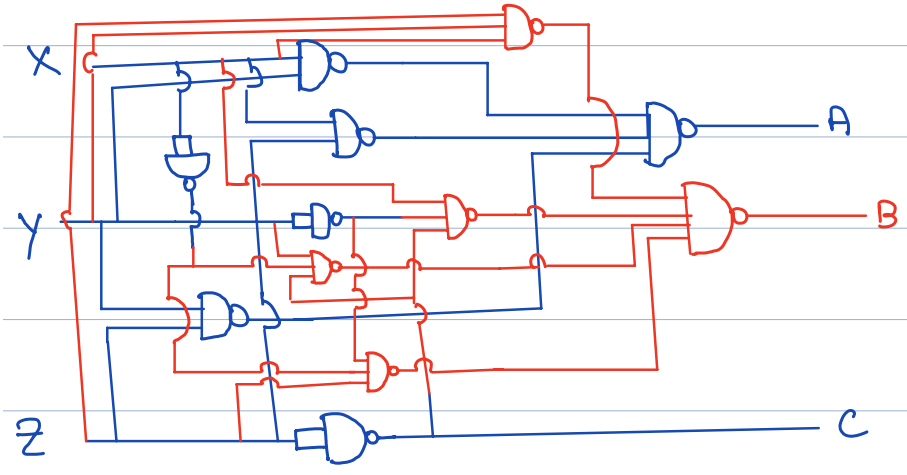
$$AB, \quad (AB)', \quad A + B, \quad (A + B)'$$

8. Implement the function $f(A, B, C) = \sum m(3, 5, 6, 7)$ using a 3-to-8 decoder and any additional logic gates (as required).

1. Consider a logic circuit with three inputs, namely x, y, and z, and three outputs, namely A, B, and C. When the binary input (i.e., xyz) is 0, 1, 2, or 3, the binary output (ABC) should be one greater than the input. For example, if $xyz = 000$, then $ABC = 001$. When the binary input is 4,5,6, or 7, the binary output should be one less than the input. Design a circuit to realize this using only NAND gates.

when $x=0 \Rightarrow$ we add 001
when $x=1 \Rightarrow$ we add 111
$\Rightarrow$ first 2 bits match with x and last one is always 1.

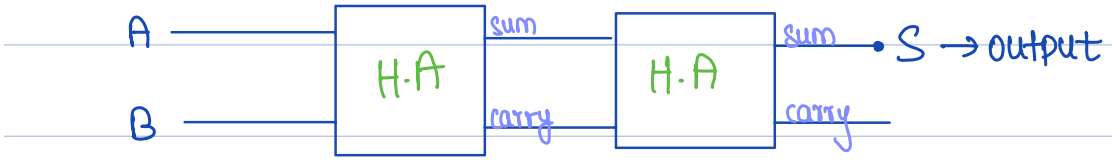




12 NAND GATES

2. A half-adder takes two inputs (each of which is a single bit number) and produces the sum and carry of addition of the two input bits. In this question, we explore the possibility of using a half-adder as a primitive. Design a 2-input OR gate using minimum number of half adders.

A	B	S	C	S'	C'
0	0	0	0	0	0
0	1	1	0	1	0
1	0	1	0	1	0
1	1	0	1	1	0



4. Consider the same binary inputs as in question 3. Assume that we have a 4-bit carry lookahead adder (instead of a ripple carry adder). Perform the addition showing the numerical values of the generate and propagate signals as well as the carry calculations using the generate and propagate equations.

$a_3a_2a_1a_0 = 1011_2, \quad b_3b_2b_1b_0 = 1101_2$

$G_i = A_i \cdot B_i$ $\quad P_i = A_i \oplus B_i$ $\quad C_i = G_{i-1} + P_{i-1} C_{i-1}$

i	Ai	Bi	Gi	Pi
0	1	1	1	0
1	1	0	0	1
2	0	1	0	1
3	1	1	1	0

$C_0 = 0$

$C_1 = 1 + 0.0 = 1$

$C_2 = 0 + 1.1 = 1$

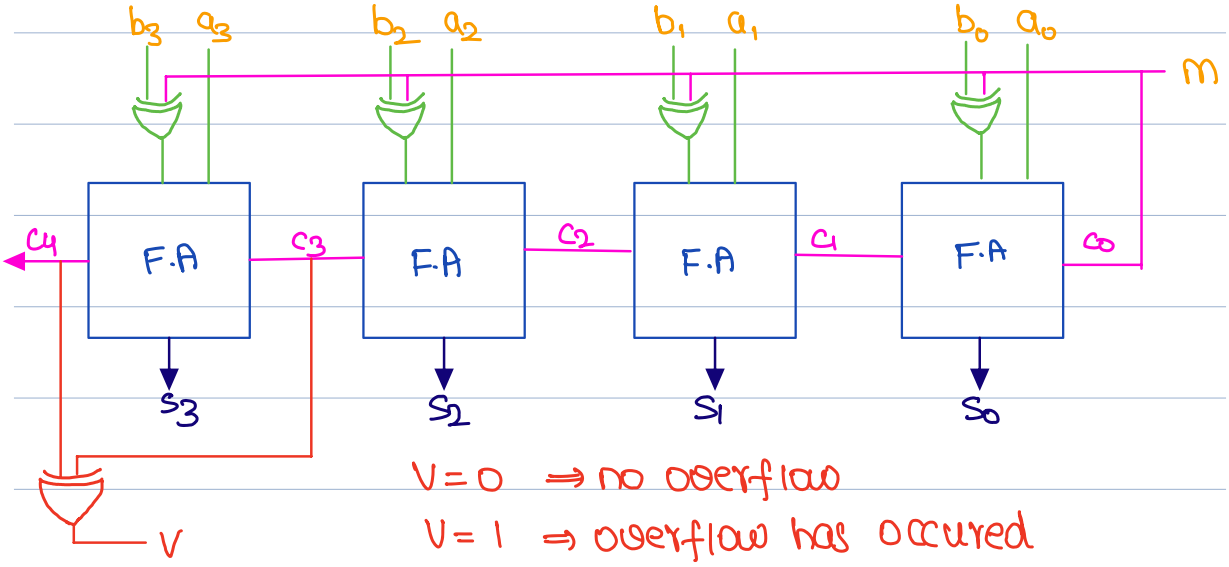
$C_3 = 0 + 1.1 = 1$

$C_4 = 1 + 0.1 = 1$

$S_i = P_i \oplus C_i \Rightarrow S_0 = 0 \quad S_1 = 0 \quad S_2 = 0 \quad S_3 = 1$

$\Rightarrow S = 10000_2$ (carry over = 1)

6. Design an adder/subtractor unit using a 4-bit binary adder and some external gates. The adder/subtractor should give out $A + B$ if $M = 0$ and $A - B$ if $M = 1$ where M is a mode bit. Also, provide an indicator for checking the overflow.

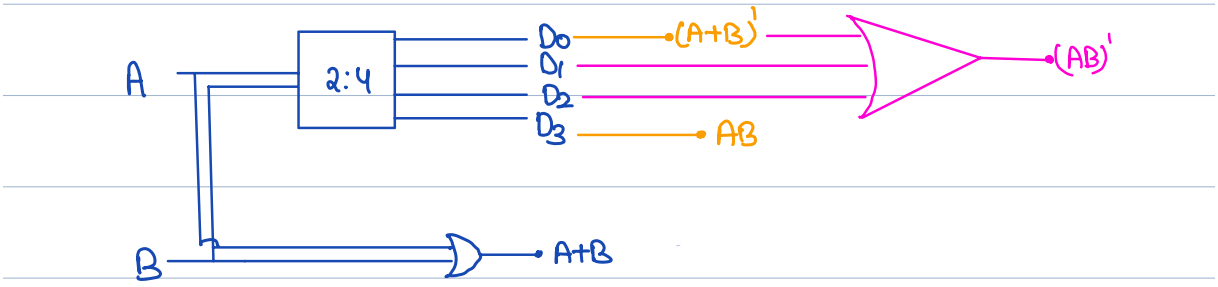


V = 0 ⇒ no overflow

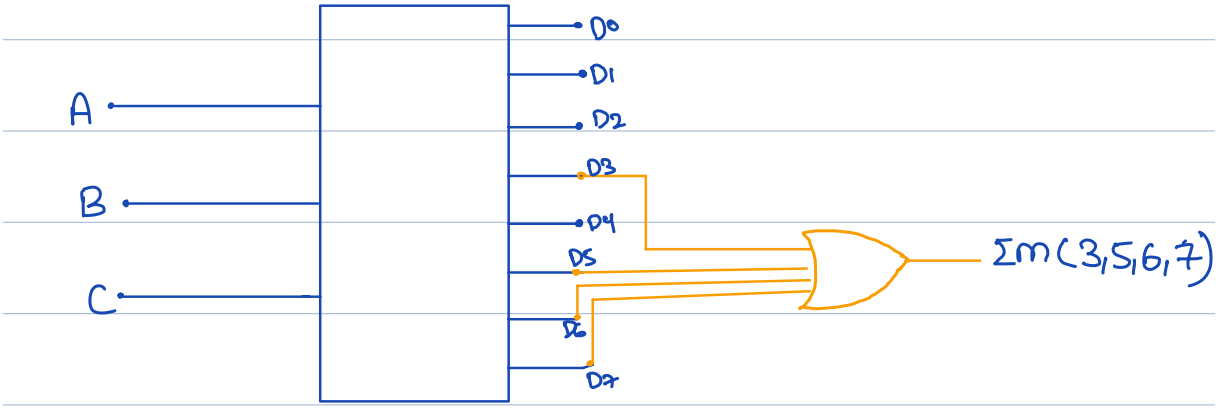
V = 1 ⇒ overflow has occured

7. You are given a 2:4 decoder, a 2-input OR gate, and a 3-input OR gate. Using these components, design a circuit that takes A and B as inputs and generates the following four outputs:

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